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RESEARCH ARTICLE

Automatic Generation of High-Quality MCQs With LLMs for Artificial Intelligence Education

CHI CHIU SO[1], (Member, IEEE), CHEUK HO LEE[2], DAN YANG[3], ANTHONY WAI KEUNG LOH[1], AND CHUEN KAN SZE[1]

[1]School of Professional Education and Executive Development, The Hong Kong Polytechnic University, Hong Kong, China
[2]Department of Computing, The Hong Kong Polytechnic University, Hong Kong, China
[3]School of Education Science, Nanjing Normal University, Nanjing 210097, China

Corresponding author: Chi Chiu So (kelvin.so@cpce-polyu.edu.hk)

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ABSTRACT Modern education demands meticulous design of teaching and learning materials to foster effective learning. Multiple-choice questions (MCQs) are widely used in education worldwide due to their high scalability, automated grading efficiency, and the ability to evaluate conceptual understanding across various topics. However, creating high-quality MCQs that match learning objectives accurately, balance different question types, Bloom's revised taxonomy levels, difficulty levels, and avoid repetition of content is demanding and time-consuming even for experienced educators. Large Language Models (LLMs) with deep reasoning capabilities offer novel opportunities for automated MCQ generation. Creating high-quality Multiple-Choice Questions (MCQs) is a time-consuming challenge for educators. This paper introduces a sophisticated pipeline that leverages Large Language Models (LLMs) to automatically generate high-quality MCQs for Artificial Intelligence education. Our method employs a zero-shot prompting strategy with different question types to guide LLMs in creating questions. To ensure reliability and mitigate inaccuracies, the pipeline integrates a novel Chain-of-Verification (CoVe) methodology called Cross-CoVe to systematically validate the generated content. The generated MCQs of our pipeline are rigorously evaluated on MCQ quality, explanation quality, and degree of diversity in a case study in an Artificial Intelligence (AI) course. We generated 200 MCQs, and the results demonstrate that 94.5% of MCQs had a single, correctly identified answer, and 79% provided robust explanations for both the correct answer and the distractors. The pipeline also achieved excellent diversity, with 99% of questions being non-duplicates. Notably, our Cross-CoVe verification strategy proved highly effective, correctly identifying 63.6% of flawed questions—a statistically significant improvement ($p = 0.002$) that more than doubles the performance of a self-verification baseline (27.3%). Our work contributes to the intense research and discussion on AI-driven educational tools, highlighting the high potential of state-of-the-art LLMs with deep reasoning capabilities to assist rather than replace educators, especially in large-scale educational resource development.

INDEX TERMS Large language models (LLMs), Bloom's revised taxonomy, chain-of-verification (CoVe), multiple-choice questions (MCQs).

I. INTRODUCTION

Modern education requires meticulous design of educational resources and proper assignment of teaching and learning activities to foster active learning and improve logical thinking. Multiple-choice questions (MCQs) are widely used in education due to their versatile scalability, automated grading efficiency, and ability to evaluate conceptual understanding across various topics. To answer an MCQ, students must understand fundamental concepts and apply

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critical thinking appropriately [1], [2]. However, creating high-quality MCQs that align with learning objectives accurately, balance different question types, Bloom's revised taxonomy levels, difficulty levels, and avoid repetition of content is demanding and time-consuming even for experienced educators. This difficulty in generating useful and high-quality MCQs is further exacerbated by the rapid evolution of the latest knowledge and skills that students should learn and the pressing need to prevent academic integrity violations given easily accessible online resources in the era of artificial intelligence and big data [3], [4].

A Typical MCQ,

Which of the following best describes the bias-variance trade-off?

A) High bias leads to overfitting, while high variance leads to underfitting.
B) Reducing model complexity decreases variance and increases bias.
C) A model with high variance performs well in training data, but poorly in unseen data.
D) Ensemble methods primarily address bias rather than variance.
E) None of the above.

However, creating high-quality MCQs that align accurately with learning objectives, balance different question types and cognitive skill levels, and avoid content repetition is a demanding and time-consuming task, even for experienced educators [5]. This difficulty is further exacerbated by the rapid evolution of knowledge and the pressing need to prevent academic integrity violations in the era of artificial intelligence and big data [3], [4]. In addition to question design, explanation of the suggested answer of the MCQ is important for student learning [6]. This is especially true when the MCQs are not for assessment purposes. This drives the demand for automated generation processes.

Recent advances in large language models (LLMs) with deep reasoning capabilities, such as DeepSeek-R1 and Grok-3, offer novel and promising opportunities to automate MCQ generation. These LLMs are pre-trained with a vast amount of publicly available data. Unlike traditional approaches which rely heavily on crowd-sourcing or template-based systems, they can operate efficiently and effectively in a fully zero-shot and single-pass generation approach, and without extra data fed by users. A zero-shot prompt approach requires only structured instructions without the need for examples, and a single-pass generation approach means that LLMs can output MCQs in a single invocation.

Existing approaches to MCQ generation, including template-based systems, crowdsourcing, and early LLM applications, have made strides but fall short in several key areas [5], [7], [8], [9], [10], [11]. For instance, prior work relying on learning objectives or reference texts generates limited questions per invocation [8], often without rigorous evaluation of quality, diversity, or explanations. In a study that used single-pass generation with an LLM to generate multiple MCQs per invocation [12], the quality of the MCQs was not rigorously evaluated and diversity was not studied.

To address these gaps, we propose a sophisticated pipeline that leverages state-of-the-art LLMs with deep reasoning capabilities to automatically generate high-quality MCQs in bulk for an AI/ML course.

Our pipeline leverages a zero-shot prompting strategy with five distinct question types: basic definitions, key characteristics, difference identification, advantages and disadvantages, and example scenarios. This approach implicitly guides the LLMs to generate questions at different levels of Bloom's revised taxonomy [13], [14], [15]. We also ask the LLMs to generate explanations for all answers, and we evaluate their quality. Furthermore, our methodology integrates the Chain of Verification (CoVe) technique to minimize the false acceptance of erroneous questions by having a second LLM verify the generated answer. This design ensures diversity, accuracy, and educational alignment, using topics and subtopics as inputs to create a high-quality question pool.

Our evaluation framework is based on the rubrics proposed by [12] to evaluate the quality of the MCQs. Furthermore, our evaluation framework goes beyond the correctness by measuring explanation quality and diversity, thus offering a more holistic assessment of MCQ effectiveness.

To investigate whether and how LLMs could generate high-quality MCQ assessments for AI/ML courses, we analyzed the following research questions:

- **RQ1:** To what extent do the generated MCQs meet typical quality requirements? In detail, we evaluate the following:
  (i) the MCQ quality;
  (ii) the quality of explanation;
  (iii) the degree of diversity achieved by LLMs;
- **RQ2:** How well is CoVe at minimizing the false acceptance of MCQs?

By carrying out this work, we provide the following contributions to the computing education research community. To the best of our knowledge, this is:

C1: A sophisticated pipeline for automatic generation of high-quality MCQs for an AI/ML course.
C2: One of the first studies to assess the quality of explanation and degree of diversity of MCQs generated by LLMs.
C3: One of the first studies to attempt to evaluate the effectiveness of CoVe in minimizing false acceptance.

II. BACKGROUND AND RELATED WORK

A. USE OF MULTIPLE-CHOICE QUESTIONS (MCQS)

Multiple-choice questions (MCQs) have been a dominant and popular assessment method across the world for over a century. Their origin is diverse in history, but it is commonly believed that they are linked to measurement of intelligence quotient (IQ), educational reform over the past century, and the need for efficient test administration [16]. Today, MCQs

are widely and popularly used due to advantages such as the ease of marking, obvious objectivity in grading, and high scalability to more content [17]. In particular, MCQs are commonly preferred by students over other question types, as MCQs are usually perceived as a fairer way of assessment [18]. In implementation, MCQs are generated and randomly selected from a large question bank with categorization of questions into different difficulty levels, to nullify the ''luck factor''.

Various researchers have focused on methodologies to improve the reliability and validity of MCQs as an assessment tool [19]. Some other researchers instead focus on studying how MCQs can enhance student learning, for example, [20] claim that MCQ-based tests can enhance long-term retention and deeper understanding; [21] found that asking students to author MCQs as a learning activity has a positive impact on their learning performance.

Although there are many ongoing studies on finding optimal practices for the use of MCQs in both assessment and learning, educators generally agree on some general principles for the design of MCQs [16], for example,

- Primarily include simple-format questions and avoid overly complex item types;
- Challenge students moderately while allowing ample chances of success;
- Target specific cognitive skills related to the learning objectives.

These principles are indicative as a general guideline, but too vague as evaluation criteria of automated MCQ generation systems like the one introduced in this paper. Therefore, more detailed rubrics are needed for evaluation, which will be further discussed in Section V.

B. BLOOM'S REVISED TAXONOMY

Bloom's revised taxonomy provides a structured framework for classifying these cognitive skills, updating the original Bloom's taxonomy, a framework established in 1956 by the educational psychologist Benjamin Bloom to classify educational objectives [13]. The revised version, introduced in 2001, reorganizes and updates the original categories to emphasize a dynamic conception of educational processes [14], [15]. The 6 categories of cognitive skills in Bloom's revised taxonomy are:

1) Remembering: Retrieving relevant knowledge from memory.
2) Understanding: Comprehending the meaning of informational materials.
3) Applying: Using knowledge in new and concrete situations.
4) Analyzing: Breaking material into its constituent parts and detecting how the parts relate to one another and to an overall structure.
5) Evaluating: Making judgments based on criteria and standards.
6) Creating: Putting elements together to form a coherent or functional whole; reorganizing elements into a new pattern or structure.

According to Bloom's revised taxonomy, these 6 categories of cognitive skills follow a cumulative hierarchical order, and higher-order cognitive skills typically correspond to cognitive skills above the level of ''understanding'' [22]. Automatically generated high-quality multiple-choice questions should be able to foster students' learning and assess their proficiencies in these higher-order cognitive skills appropriately.

Therefore, an ideal MCQ generation system should generate MCQs with a balanced diversity of taxonomy levels; such a requirement will be used as evaluation criteria for our own methodology in Section V as well.

C. EXISTING METHODS OF AUTOMATED MCQ GENERATION

Designing well-structured questions and answer choices for effective teaching and learning is a meticulous and time-intensive task, even for highly experienced educators. Hence, researchers have focused heavily on the task of automated question generation. Most work in this field focuses on a specific MCQ element, i.e. the stem (question), the key (correct answer), and the distractors (incorrect options).

The most researched task is question answering (QA) [23], [24]; this is equivalent to key generation. The generation of questions (QG) focuses on reading comprehension tasks, relying on both traditional NLP methods [25] and neural networks [26] to generate meaningful questions [27]. More recent systems rely on sequence models [28] or attention-based methods such as LLMs [29], [30], [31], [32]. In addition, there have been attempts to tackle the QA and QG tasks jointly [33]. Current work on automated distractor generation (DG) focuses on cloze tests [34], [35], [36] and reading comprehension [37], [38], as well as MCQs for domains outside of computing such as biology [39]. These systems use the stem and/or the key to generate plausible (i.e., similar) distractors. Often, systems generate several more distractors than necessary and select the best through a ranking system [9].

There are different methods to generate high-quality MCQs:

1) GENERATED MCQ BASED ON REFERENCE TEXT

In [10], the authors propose an end-to-end pipeline that generates all three elements of an MCQ, with QG, QA and DG, using the fine-tuned Google's T5 transformer model on the DG-RACE dataset. They also note that evaluation is difficult and that they evaluated only 10 questions. Traditionally, most pipelines generate MCQs based on the reference text supplied.

In [11], the authors adopt a single-pass generation approach using ChatGPT Plus to generate MCQs. They rely on medical textbooks to provide context.

The number of MCQs generated depends on user input or is as many as possible. In the latter case, a long input paragraph generally generates more questions [40]. An input subject and difficulty level could be used to refine the output of the LLM [8].

This method has limited generalizability; the model can only generate MCQs based on an existing database, and this limitation is further exacerbated by the rapid evolution of the latest knowledge and skills that students should learn.

2) MCQ GENERATION BASED ON PROMPTING TECHNIQUES

Recently, LLMs have been pre-trained on vast public data and equipped with deep reasoning capabilities. Deep reasoning capability enables zero-shot prompt design and a single-pass generation approach. New pipelines have been designed without providing reference text for LLMs. Nasution [7] uses the single-pass generation approach to generate MCQs in higher education biology without relying on a reference text.

In [12], the authors designed a pipeline that generates MCQs from learning objectives (LOs). Each LO is classified as a unique level according to Bloom's revised taxonomy using a binary BERT classifier [41] and further mapped to one or many question types:

1) Recall – Given minimal to no code, students are asked about basic programming concepts or technical details.
2) Fill in the Blank – Given a code snippet with some sections removed, students are asked to select an option that successfully replaces the blank to create syntactically and semantically correct code.
3) Scenario Based – Given a scenario or situation, students are asked to identify appropriate tools, methods, or packages best suited to accomplish the task prescribed.
4) Correct Output – Given a code snippet, students are asked to trace program execution to determine either intermediate or final outputs.
5) Code Analysis – Given a code snippet, students are asked to spot errors or to build on or use the code in a new manner.

For each such LO and question type combination, an MCQ will be generated. The results are evaluated according to the following rubrics:

1) Does the MCQ provide enough information to arrive at an answer?
2) Is there a correct answer listed in the options? Is the option marked ''correct'' actually correct?
3) Are the options distinct from each other, ensuring they are unique choices?
4) Is the MCQ free from obviously-wrong options?
5) If code is present in the question prompt OR in the correct answer, is it syntactically and logically correct?
6) Does correctly answering the MCQ help in achieving the learning objectives?

Although generated MCQs are carefully evaluated and compared with human-crafted ones, the generated explanations are not evaluated.

Since module LOs restrict the output of MCQs, we used topics and subtopics instead. Module LOs are good for specific courses and not suitable for the MCQ question pool, which will be shared by courses throughout the years and used by different teachers.

Rather than strictly adhering to frameworks like Bloom's revised taxonomy, teacher questioning is a dynamic practice aimed at assessing knowledge, promoting comprehension, and stimulating critical thinking from multiple angles [42], [43].

In [44], the authors propose a framework that generates MCQs. It integrates an LLM with retrieval-augmented generation (RAG) and prompt engineering techniques. This ensures that the generated questions are factually accurate and contextually relevant by dynamically pulling information from a designated knowledge base during the generation process. However, these generated questions do not always align with the actual intent or accuracy required in educational settings.

To address these gaps, we propose a new pipeline for generating an MCQ question pool using zero-shot prompt design, a single-pass approach, and no reference text. This pipeline is distinguished by several key contributions. First, by using broader topics and subtopics instead of LOs, we create more versatile and reusable question types and prompt templates to be domain-agnostic and suitable for a wide range of courses.

More importantly, our work is among the first to conduct a holistic evaluation that rigorously assesses not only the quality of the MCQs themselves but also the quality of the generated explanations and the degree of diversity.

Finally, to enhance reliability, we integrate and evaluate the CoVe technique to systematically filter out erroneous questions, an aspect not thoroughly explored in previous studies.

III. OUR METHODOLOGY

In this section, we detail the sophisticated pipeline developed for automated generation of a high-quality MCQ question pool. Our novel approach leverages state-of-the-art LLMs with zero-shot prompting strategy, which requires no examples, and a single-pass generation approach, enabling LLMs to produce MCQs in a single invocation. This approach produces high-quality question pools from course topics and subtopics, without requiring reference texts or fine-tuning. This approach ensures scalability for large-scale educational resource development. Our pipeline integrates CoVe verification to ensure the accuracy and reliability of the generated content and to reduce LLM hallucinations. The entire workflow is illustrated in Figure 2.

The generated MCQs will be evaluated with rubrics similar to those used in Jacob Doughty [12]. We also introduce additional rubrics to measure the quality of the solution generated by LLMs.

We will present and discuss each of these features one by one in the following.

A. GENERATOR LLMS: DEEPSEEK-R1 AND GROK-3

DeepSeek-R1 and Grok-3 are state-of-the-art LLMs with strong deep-reasoning capabilities. In our novel methodology, DeepSeek-R1 and Grok-3 will be used as LLM engines to automatically generate MCQs in the zero-shot prompting approach. We call them *generator LLMs*. Below are some brief descriptions of the two generator LLMs.

DeepSeek-R1 is a large language model (LLM) released in early 2025 and is renowned for its extraordinarily low training cost and outstanding reasoning performance enabled by reinforcement learning techniques in the training process [45], [46]. DeepSeek-R1 distinguishes itself as the first LLM to integrate several pioneering techniques which contribute to the enhanced reasoning capabilities and cost efficiency, such as the use of Mixture of Experts (MoE) architecture and the Group Relative Policy Optimization (GRPO) algorithm.

Grok-3, unveiled in February 2025 by xAI, has been recognized for its advanced capabilities in various domains, particularly in real-time processing tasks. Grok-3 has even been reported to outperform DeepSeek-R1 and other state-of-the-art LLMs [47]. However, some researchers suggest that while Grok-3 offers advantages in specific domains, DeepSeek-R1 may provide more reliable outputs in scenarios that require rigorous logical reasoning and structured problem solving. Therefore, it is intriguing for us to investigate the reasoning capabilities of both LLMs and explore how useful they are in the automated generation of MCQs in computing education.

These LLMs process the complex instructions in the prompt to generate a batch of MCQs that adhere to the specified guidelines.

B. FIVE QUESTION TYPES

Instead of using assessment formats such as ''Recall'', ''Fill in the Blank'', etc. as in [12], which are suitable for programming courses. We focus on generating a high-quality MCQ question pool, with question types designed to produce questions from different perspectives and cognitive levels. To achieve it, we designed different question types ''Basic Definitions'', ''Key Characteristics'', ''Difference Identification'', ''Advantages and Disadvantages Comparison'' and ''Example Scenarios''.

C. OUR PROMPTING STRATEGY

Prompt engineering is the development of input prompts strategically to tailor LLMs for specialized applications such as MCQ generation. This technique involves constructing precise textual inputs to steer the model's generative process toward specific, desired outcomes. We leverage prompt engineering to direct the LLM to generate questions in different question types that can enhance and challenge students. By skillfully designing prompts, the LLM is directed to leverage its extensive parametric knowledge, thereby generating questions of high pedagogical value and contextual relevance. An effective prompt formulation constitutes a critical step in constraining the generative process of LLM. The technique directs the generative output of LLMs to construct MCQs that are aligned with established pedagogical criteria for question formulation. Our prompting strategy adopts a zero-shot prompt, single-pass generation approach, and no reference text. LLMs with deep reasoning capabilities particularly excel at completing complex human instructions; therefore, no additional resources are required, except the basic prompt template.

To ensure comprehensive comparison between subtopics, prompts have to be designed to target one core topic (for example, neural networks and searching in an introductory AI/ML course) and multiple subtopics (for example, A* searching algorithm and convolutional neural networks in an introductory AI/ML course). The core topic and subtopics are both used as keywords and will be systematically embedded into our prompt templates to guide topic selection in the MCQ generation process. In other cases, we will let the LLM determine the subtopics for comparison and no subtopics will be provided. We have another template for this case, which will be discussed in the next section III-D.

Our prompting strategy has 3 key components:

1) Introducing roles and specifying topics and subtopics.
2) Providing explicit requirements with respect to the question format and output format.
3) Setting detailed guidelines on the automated MCQ generation process.

These three parts are discussed in-depth below.

1) For the first part, we introduce to the LLM its identity as a teacher who is responsible for generating certain amounts of MCQs for specific topics and subtopics in a specified course with course description. An example prompt for part 1 is:
   Prompt Example - Part 1
   You are a teacher of course { Introduction to Artificial Intelligence and Machine Learning }. A multiple-choice question is a question with one correct answer and multiple distractors. The question and the correct answer are also called stem and key, respectively. Please generate { 25 } multiple-choice questions for the core topic { Searching } with subtopics: { Uniform-cost search, Greedy best-first search, and A* search }.
   Prior Knowledge: { Calculus, probability and statistics, data structures and algorithms }
   Course description: { This course aims to equip students with the main concepts, ideas and techniques of artificial intelligence (AI) including machine learning. It covers the basic building blocks of AI and its applications in different domains. This subject also aims to enable students to gain practical experiences in applying the techniques in AI to solve real-world problems. }
   The number of MCQs and the choice of core topic and subtopics can be changed according to the specific

needs in the real use cases. We take 25 MCQs and some searching algorithms here as an illustrative example for an introductory AI/ML course.

2) In the second part, we require the generated MCQs to be in certain formats with five choices and the number of choices can be changed. If the question is related to graphs or neural networks, the question must first clearly define the graphs or neural networks as background information before presenting the exact question. In particular, the correct answer must be specified, along with a detailed explanation of the correct answer.

   Prompt Example - Part 2

   Questions should be formatted in the following styles:

   (1) Each question should have { 5 } choices, where one and only one key, others are distractors. Label choices with letter A), B), C), ...
   (2) Define graphs or neural networks at the beginning of the questions if they are needed.
   (3) Correct answers should be given, along with detailed explanations including a step-by-step deduction of the answers.
   (4) The output should be in json format including the keys ''question'', ''choices'', ''correctanswer'', ''explanation''.

3) In the third part, we present some detailed guidelines for LLMs to follow, which are summarized in 8 key steps. Two of the most important steps include asking LLMs to understand the topics and subtopics in a comprehensive way and design the questions from 5 question types and discussed in section III-B, namely, ''Basic Definitions'', ''Key Characteristics'', ''Difference Identification'', ''Advantages and Disadvantages Comparison'' and ''Example Scenarios''. Other important steps include asking the LLMs to generate questions with a balanced diversity of difficulty levels, without any ambiguity and biased language. In addition, questions should be set to test the various forms of misconceptions among students.

   Instruction (4) has been embedded in the prompt to lead the LLMs to generate more ''Creating'' questions. A detailed prompt example of part 3 is presented below:

   Prompt Example - Part 3

   Follow these generation guidelines:

   (1) Understand the topic and subtopics: First, make sure you have a solid understanding of the core topic and subtopics and the related concepts, definitions, key properties, similarity and differences, and applications.
   (2) Generate question with the following question types:
      a. Basic Definitions
      b. Key Characteristics
      c. Difference Identification
      d. Advantages and Disadvantages Comparison
      e. Example Scenarios
   (3) Try to create distractors with common misconceptions.
   (4) Try to create a few questions to test the creativity of students by either presenting a new, unexpected scenario and asking students to apply their learning creatively to solve a problem or meet a need, or posing counterfactual or speculative scenarios to spark imagination and requiring students to apply logic to unlikely situations.
   (5) Balancing Questions: Ensure a balanced mix of easy, medium, and hard questions with different question types to cater to different levels of understanding.
   (6) Avoid Ambiguity: Make sure each question has one and only one clear correct answer to avoid confusion.
   (7) Neutral Distractor: Distractors should be different, and neither be obviously wrong nor give clues or hints to the correct answer.
   (8) Check for Bias: Ensure that the questions are neutral and do not inadvertently suggest the correct answer through wording.

Again, the above example is intended for an introductory course in AI/ML. The complete prompt template of this illustrative example is provided in Appendix A for interested readers. The experimental results of this prompt template with different LLMs will be presented, evaluated and discussed in section V.

D. FURTHER AUTOMATION - WHEN SUBTOPICS ARE UNNECESSARY

In the prompt template introduced in Section III-C, the prompt asks for the generation of MCQs on a few related subtopics together, such as Uniform-cost search, Greedy best-first search and A* search. Therefore, questions around these related subtopics can be generated, for example, questions that compare different search algorithms by their key characteristics, advantages, and disadvantages.

Another advantage of specifying subtopics is that they restrict the scope of a topic. However, sometimes we do not want to restrict the content of a topic. In this case, the LLMs will need to be able to discover relevant subtopics by their own knowledge and reasoning. For this purpose, we will need another prompt different from the one presented in section III-C and appendix A.

Hence, we provide two prompt templates. The first one, as shown in section III-C and appendix A, is for user-specified subtopics. The second comes with further automation to allow LLMs to decide which subtopics should be included and can be found in the Appendix B. It is constructed with minor changes from the first prompt template for automation of subtopic suggestions. Some examples using the two templates with and without user-specified subtopics in an introductory AI/ML course are presented in Table 2.

E. CHAIN-OF-VERIFICATION (COVE)

In the above, we have presented our detailed prompting strategy for the automated generation of MCQs. However, LLMs can make careless mistakes just as humans. It would be helpful to filter out questions with incorrect suggested answers. With the objective of minimizing false acceptance, an advanced technique, Chain-of-Verification (CoVe) [48], [49], is used to improve the overall quality and reliability of the generated MCQs. After the generation of MCQs by DeepSeek-R1 or Grok-3, a new LLM will be prompted to double-check the correct answer. That new LLM is called a *verifier LLM*.

We use the following prompt template for CoVe:

Prompt Example for CoVe

You are a teacher of an Artificial Intelligence and Machine Learning course. You are given the following multiple-choice question (MCQ).

The MCQ: { The MCQ }

What is the correct answer to this MCQ?

Select from 5 options: A / B / C / D / E.

For each generated MCQ, the above prompt will be sent to the verifier LLM three times, and by using the majority vote, a final answer will be selected as the verifier answer. If the verifier answer is different from the answer suggested by the generator LLM, the MCQ will be marked as a failure in the CoVe test.

In our experimental configuration, we set up two CoVe cases: Normal-CoVe refers to self-verification, where the generator LLM also acts as the verifier, while Cross-CoVe employs a different LLM as the verifier. The accuracy represents the true rejection rate for questions with wrong answers.

The Normal-CoVe and Cross-CoVe settings are shown in Figure 1.

The results show that Cross-CoVe, with an accuracy rate of 63.63%, performs much better than Normal-CoVe at 27.27%. This result illustrates that the Cross-CoVe verification scheme has better distinguishing ability.

All in all, our prompting strategy can be summarized in a generation cycle with a structured workflow:

1) If there are user-specified subtopics, go for the prompt template in appendix A;
2) If there are no user-specified subtopics, go for the prompt template in appendix B;
3) Fill in the selected prompt templates;
4) Send the completed prompts to the generator LLMs;
5) The generator LLMs will then return a certain number of candidate MCQs for each prompt;
6) The generated MCQs are fed to a verifier LLM to review the correctness of the generated MCQs in the framework of CoVe.

The prompting strategy is shown in Figure 2.

IV. EXPERIMENTAL DESIGN

To evaluate the quality of the automatically generated MCQs, we generate 25 MCQs for each topic and a total number of 200 MCQs as in Table 1. We use DeepSeek-R1 with default temperature 0.7 and Grok-3 with default temperature 1. Default temperatures are set to balance creativity and coherence in responses. Depending on the existence of subtopics, different prompt templates are used in the generation of MCQs as in Table 2.

We refer to the rubrics used in [12] to evaluate the quality of MCQs except those specially designed for programming courses. In addition, we add more to measure the quality of the explanation and the diversity of MCQs. Rubrics on question types, difficulty levels, Bloom's taxonomy levels, duplication, and computation will be used to measure the diversity of MCQs generated. These qualities are important for MCQs generated in bulk and are measured and reported here for the first time.

The first 5 rubrics are used to evaluate the quality of the MCQs as in Figure 3.

Rubric 6 evaluates the quality of the explanation provided by LLM as shown in Figure 4.

Rubrics 7 to 11 evaluate the diversity of MCQs generated as shown in Figure 5.

For the evaluation of MCQs, we employ five expert annotators to evaluate the quality, explanation, and diversity of MCQs. Each annotator was tasked with evaluating 200 MCQs using 11 rubrics. This will make the evaluation result more consistent, especially when evaluating the diversity quality. Five instructor annotators applied the rubrics, 1000 annotations for the 200 MCQs were produced. We required annotators to attempt to answer the question before applying the rubric items. In addition, annotators must complete the evaluation of all questions within a topic before answering Question 10, so that they can capture duplicated questions. We do not compare duplication of questions generated by different LLMs. To ensure the reliability and integrity of the annotation data, this study employs expert annotators, as the rubric's inclusion of specialized concepts like Bloom's revised taxonomy levels would likely lead to inconsistent application by student cohort.

To evaluate the reliability of these annotations, we measure interrater agreement. A conventional metric for this purpose is Fleiss's kappa [50]. However, kappa is known to severely underestimate the agreement in situations when one of the labels is clearly dominant [51]. This statistical artifact is directly pertinent to our dataset, where for rubric items 1∼ 6 and 10∼ 11, there is always a clearly dominant answer. To address this issue, we compute and use Gwet's AC1 which is robust in such situations [52]. Table 5 reports the AC1 and the scores for each of the eleven rubric items. The measured Gwet's AC1 ranges from 0.69 to 0.99. We can see that raters diverge on difficulty levels and Bloom's revised taxonomy levels. Each MCQ was annotated by all instructors. To resolve disagreements, we used the following rules in the presented order:

1) Majority Vote;
2) An annotator who correctly answered the MCQ had precedence over the one who answered incorrectly;

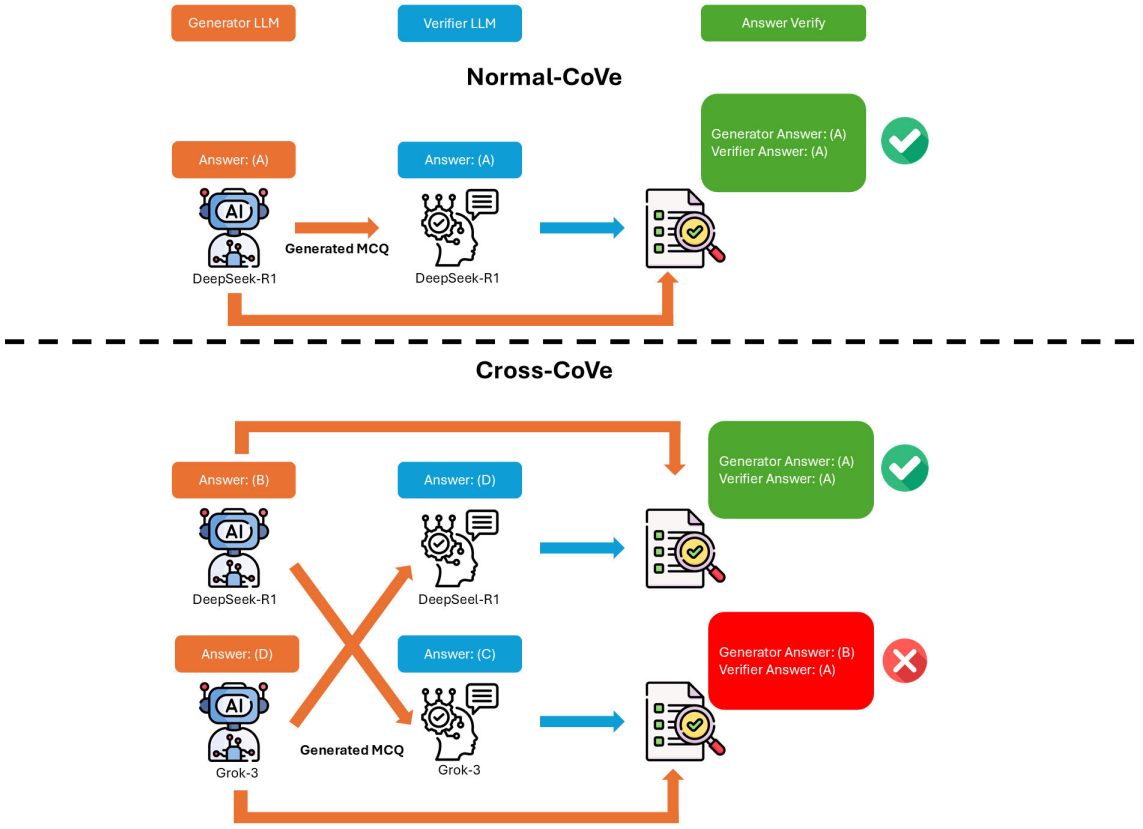


FIGURE 1. An illustration of Normal-CoVe and Cross-CoVe.

TABLE 1. Number of MCQs generated by each LLM.

Topic	LLM		
	DeepSeek-R1	Grok-3	Total
Searching	25	25	50
K-means Clustering	25	25	50
Neural network	25	25	50
Naive Bayes classifier	25	25	50
Total	100	100	200

3) The least favorable evaluation;

To evaluate the effectiveness of CoVe, we send the generated MCQ to the verifier LLM to prompt for an answer three times. With a majority vote, the CoVe answer will be determined. We evaluate both Normal-CoVe and Cross-CoVe.

Then, the generated suggested answer, the CoVe answer, and the correct answer marked by the annotators are compared to evaluate the effectiveness of CoVe. To test for statistically significant differences we employ Fisher's exact test. Our results show that Cross-CoVe performs much better than Normal-CoVe.

V. EXPERIMENTAL FINDINGS

A. QUALITY OF MCQ

Figure 3 shows the quality of MCQs generated. We have compared our findings with those in [12] which studies the automatically generated MCQs in programming courses. There are two notable issues with the automatically generated MCQs as mentioned in [12]. One notable issue is the multiple correct answer choices (4.9% vs 1.1% human). Our method yielded a rate of 2.50%, a notable improvement over the 4.9% reported in [12], although this is still higher than the 1.1% for human-crafted MCQs. Another issue of automatically generated MCQs is having distractors that gave away the correct answer (4.0% vs 0.9% human). We have 1.00%, which is comparable to human-generated MCQs. In summary, our automatically generated MCQ quality is better than that in [12] and comparable to the quality of human-crafted MCQs.

B. QUALITY OF EXPLANATION

Figure 4 shows the quality of the explanation for the correct answer to the MCQs provided by LLMs. We notice that there are 17.50% explanations that do not correctly explain

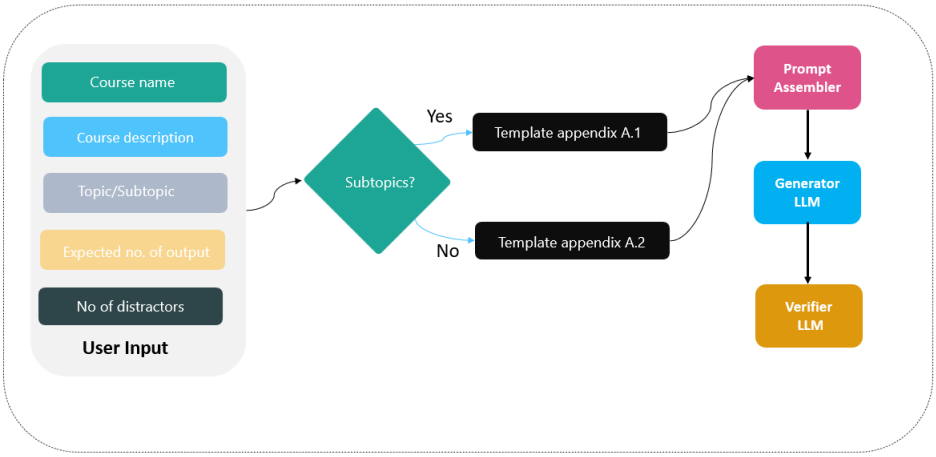


FIGURE 2. An illustration of Prompting strategy.

TABLE 2. Examples of topics, subtopics and prompt templates.

Topics and Prompt Templates		
Core Topic	Subtopic	Prompt Template
Searching	Uniform-cost search,Greedy best-first search, A* Search,	appendix A
K-means Clustering	*Not user-specified*	appendix B
Neural Networks	Convolutional neural networks, Fully-connected neural networks	appendix A
Naïve Bayes Classifier	*Not user-specified*	appendix B

TABLE 3. Inter-rater Agreement. The table shows Gwet's AC1 scores for each of the eleven rubric items.

No.	Rubric	Category	Gwet's AC1
1	Enough information	MCQ quality	0.975
2	Correct answer listed	MCQ quality	0.938
3	Distinct options	MCQ quality	0.99
4	No obviously wrong answer	MCQ quality	0.938
5	Align with topic	MCQ quality	0.988
6	Correct explanation	Explanation quality	0.788
7	Question type	Diversity	0.870
8	Difficulty level	Diversity	0.713
9	Bloom's revised taxonomy level	Diversity	0.696
10	Duplicate question	Diversity	0.951
11	Computation problem	Diversity	0.859

the wrong options. LLMs either give an overly simplistic explanation for the wrong options or simply do not mention it at all. In fact, 96.50% of the explanations correctly explain the correct answer. The overall quality of the explanations is high for the correct answers, but improvement is needed for the explanations of wrong options.

C. DIVERSITY ANALYSIS

Figure 5 shows the diversity of MCQs. We have asked the LLMs in prompt to balance the difficulty levels and Bloom's revised taxonomy levels. The results indicate that the balancing is effective. There seem to be too many questions of type "Key Characteristics" (39.00%), at the difficulty level "Medium" (61.00%), and at the taxonomy level "Understanding" (29.50%). However, even for human instructors, they will tend to create more questions about "Key Characteristics" to test the "Understanding" of students. The "Medium" difficulty level was the most frequently assigned category by annotators.

99% of the MCQs are not duplicates of another question on the same topic. However, there are many questions with similar distractors. For example, the following two questions share the same distractors. However, they are considered different questions as long as they are assessing different knowledge points from different perspectives.

Q1 Which search algorithm is most suitable for finding the shortest path in a weighted graph without heuristic information?
Q2 Which search algorithm is guaranteed to find the optimal solution if the heuristic is admissible and consistent?

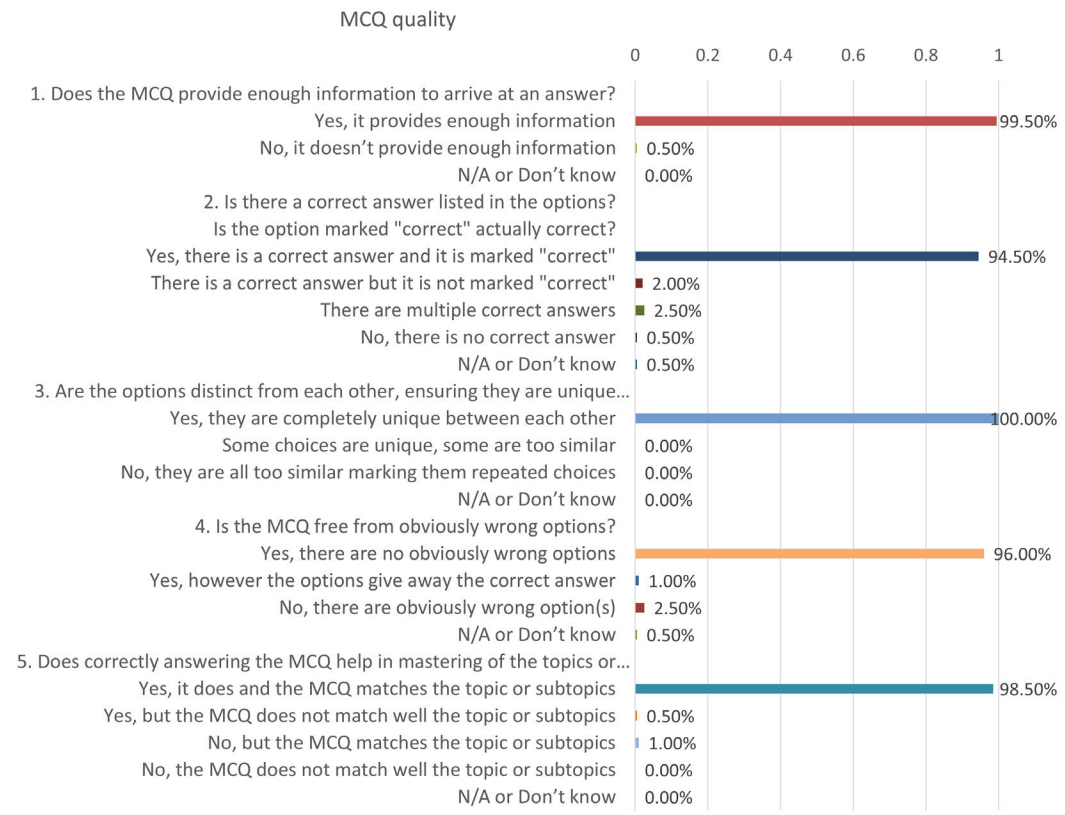


FIGURE 3. Post-Annotation rubrics results for MCQ quality.

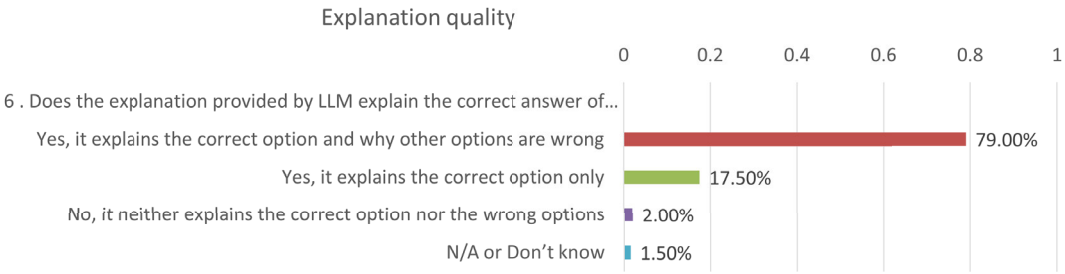


FIGURE 4. Post-Annotation rubrics results for MCQ explanation.

A) Greedy best-first search.
B) Depth-first search.
C) Breadth-first search.
D) Uniform-cost search.
E) A* search.

There are 10% of the questions that are computation questions. We did not set any requirement in the prompt. For instructors who prefer more computation questions, they may need to adjust the prompt to generate them.

D. EFFECTIVENESS OF CHAIN-OF-VERIFICATION (COVE)

Our experimental results on the introductory AI/ML course show that Chain-of-verification (CoVe) is an effective measure for filtering problematic MCQs. Table 4 presents the false acceptance (Type I error), false rejection (Type II error), and accuracy rates achieved by Normal-CoVe and Cross-CoVe.

Cross-CoVe exhibits massive performance enhancement, achieving 63.64% accuracy in filtering questions with wrong answers, more than doubling Normal-CoVe's 27.27%. At the same time, while preserving 98.41% of questions with correct answers. This improvement is especially pronounced in directional verification when DeepSeek-R1 verifies Grok-3 generated MCQs, the accuracy surges to 75%.

In contrast, Normal-CoVe delivers lower accuracy rates with 33.33% for DeepSeek as the generator and 25% for Grok-3 as the generator.

We performed Fisher's exact test for statistical significance between CoVe and without CoVe. With ($p = 0.002$) there is a statistically significant difference.

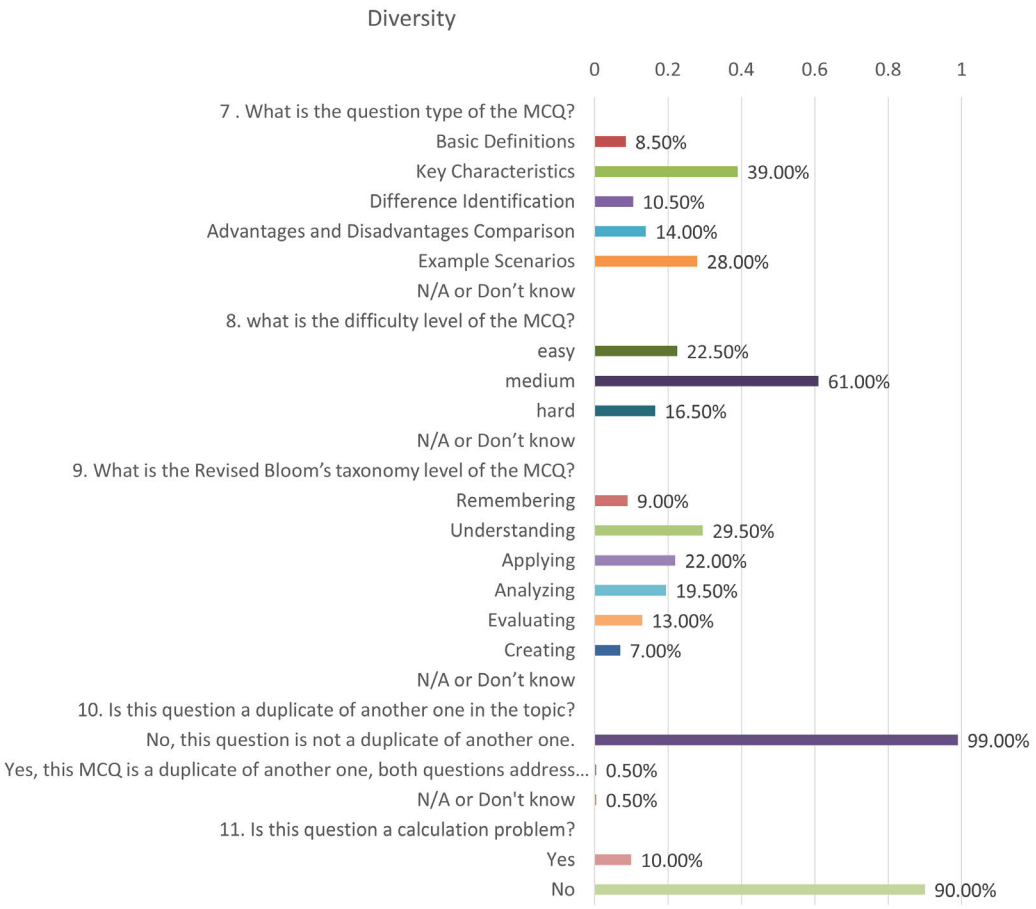


FIGURE 5. Post-Annotation rubrics results for MCQ diversity.

TABLE 4. The Effectiveness of CoVe to filter wrong questions.

LLM	Accuracy	Type I error	Type II error
Normal-CoVe			
DeepSeek-R1	1 / 3 (33.33%)	2 / 3 (66.67%)	2 / 97 (2.06%)
Grok-3	2 / 8 (25.00%)	6 / 8 (75.00%)	3 / 92 (3.26%)
Cross-CoVe			
DeepSeek-R1	1 / 3 (33.33%)	2 / 3 (66.67%)	1 / 97 (1.03%)
Grok-3	6 / 8 (75.00%)	2 / 8 (25.00%)	2 / 92 (2.17%)

In summary, we found that Cross-CoVe is an effective way to reduce the false acceptance of generated MCQs. However, further improvements are necessary to reduce the rate of false acceptance.

E. ALIGNING QUESTION TYPES WITH BLOOM'S REVISED TAXONOMY

Although there is no strict relation between the five question types and Bloom's taxonomy levels, each question type tends to align with certain Bloom's revised taxonomy levels more frequently, as shown in Table 5. For example, the question type "Difference Identification" is less likely to be assigned to "Remembering", and more likely to be categorized as "Analyzing". The actual distribution of generated MCQs per Bloom's taxonomy level and question type is shown with Table 6.

Theoretically, this approach resembles the way that human teachers create questions. Teachers do not start with Bloom's revised taxonomy in mind; instead, they craft questions based on expected student learning from various angles, such as perspectives, examples, comparisons, and applications, etc. In other words, teachers design questions from very concrete question types while considering desired difficulty levels. Eventually, different question types help generate questions that train and foster different levels of cognitive skills and allow a variety of difficulty levels for learning and assessment purposes.

VI. DISCUSSION

A. OVERALL QUALITY OF GENERATED MCQs

Our experiments show that the proposed pipeline, using state-of-the-art LLMs like DeepSeek-R1 and Grok-3, produces MCQs of superior quality compared to those in [12]

TABLE 5. Mapping question types to bloom's revised taxonomy by annotators.

Question Type	Bloom's Revised Taxonomy by Annotators
Basic Definitions	Remembering, Understanding
Key Characteristics	Understanding
Difference Identification	Analyzing
Advantages and Disadvantages	Analyzing, Evaluating
Example Scenarios	Applying, Creating

TABLE 6. Distribution of MCQs generated per question type and Bloom's revised taxonomy level by annotators.

Question Type	RMB	UND	APP	ANL	EVL	CRT	Total	Total%
Basic Definitions	11	6					17	8.50%
Key Characteristics	6	49	6	13	2	1	77	38.50%
Difference Identification	1	1		18	1		21	10.50%
Advantages and Disadvantages		3		5	20		28	14.00%
Example Scenarios			38	3	3	13	57	28.50%
Total	18	59	44	39	26	14	200	
Total%	9.00%	29.50%	22.00%	19.50%	13.00%	7.00%		100.00%

and comparable to human-crafted ones. This underscores the potential of advanced LLMs to streamline educational resource creation, particularly in dynamic fields like AI/ML, where rapid knowledge evolution demands efficient, high-fidelity question generation. To unpack these outcomes, we explore the factors driving this performance and their implications for computing education.

The pipeline's strengths in MCQ quality stem from its sophisticated prompting strategy, which embeds explicit guidelines for diversity in question types. This causal linkage ensures generated MCQs are not only syntactically precise and conceptually accurate but also free from obvious distractors, achieving 100% distinct options, compared with 98.2% by Jacob Doughty [12]. The difference may arise from the AI/ML domain's emphasis on conceptual clarity rather than code equivalence, where LLMs can more reliably discern semantic distinctions without conflating functionally similar implementations. For MCQs generated for programming courses, various program codes could lead to the same result, and the LLM cannot recognize that they are equivalent. Similarly, the overall quality of explanations is robust, providing step-by-step deductions that enhance student learning, as LLMs draw on their pre-trained knowledge to articulate reasoning effectively. However, the explanations provided for incorrect options are sometimes inadequate. Grok 3 performs better than DeepSeek-R1 in this perspective, with 95% of questions explaining both correct and wrong options properly, compared to 61% for DeepSeek-R1. This discrepancy likely reflects variations in the models' reasoning depths, with Grok-3 exhibiting stronger handling of nuanced misconceptions. To address this, future iterations might incorporate chain-of-thought prompting to enforce balanced explanations across all options or verification with human oversight, ensuring distractors are not only plausible but fully dissected for pedagogical value.

Additionally, the diversification across question types, difficulty levels, and Bloom's revised taxonomy levels is well balanced with no duplicate questions. However, the same knowledge points could be questioned several times from different points of view. This finding suggests that manual curation of generated questions may be necessary to ensure broad coverage of course content in assessments. It may suggest opportunities for algorithmic filtering to optimize content coverage.

By generating diverse MCQs that implicitly span cognitive hierarchies, from remembering to creating, the approach fosters deeper conceptual understanding and adaptability to real-world applications, such as scenario-based questions that test creative problem solving. This scalability bridges the gap between large enrollment courses and targeted learning needs, allowing instructors to curate subsets for broader topic coverage. Ultimately, the pipeline advocates for LLMs as collaborative aids in education, amplifying human expertise while promoting equitable access to high-quality resources in evolving technical domains.

VII. LIMITATIONS AND FUTURE DIRECTIONS

The quality of MCQs in this work was compared with the quality of MCQs automatically generated and human-crafted for the programming course in [12]. There could be a difference between the programming course and the AI/ML course. In addition, we do not have a reference for comparison of the quality of the explanation and the degree of diversity. Since we did not invite student annotators, some metrics like difficulty level could be biased toward the instructors' point of view.

Although our current pipeline demonstrates significant promise in the automated generation of high-quality MCQs, we propose some possible directions for further improvement. Three directions aim to broaden the system's scope,

deepen its content awareness, and strengthen its verification mechanisms.

1) GENERALIZABILITY TO OTHER COURSES

We have developed a pipeline to automatically generate MCQs for the AI/ML course, and the quality of the MCQs generated is high. However, empirical validation across diverse subject domains is necessary to confirm generalizability and robustness beyond computing education.

2) INCORPORATING COURSE-SPECIFIC CONTENT FOR ADVANCED TOPICS

We do not provide any course content for LLMs to generate MCQs. However, we may need to do so for advanced courses where those course contents may not be publicly available and not included in the pre-training of LLMs.

3) EXPLORING ADVANCED VERIFICATION METHODS

Another promising direction is to explore alternative methods to improve the quality of generated MCQs beyond CoVe. For instance, techniques such as chain-of-thought verification, in which LLM explains its reasoning step-by-step before giving a final answer. This method could be investigated to further reduce hallucinations and ensure higher reliability in the verification process.

4) EXTENDING TO ADDITIONAL ASSESSMENT FORMATS

In addition, while our current pipeline focuses on MCQs, it could be extended to generate short-answer questions. This would broaden the applicability of the system. This could involve integrating prompts that guide LLMs to produce concise, open-ended queries aligned with the same topics, subtopics, and cognitive levels.

VIII. CONCLUSION

Our work offers a novel methodology for harnessing the potential of state-of-the-art LLMs in the automated generation of MCQs for computing education. We have generated MCQs with high quality and diversity, and the explanations accompanying the correct answers are immediately usable for instructional purposes. We have a rigorous evaluation of the quality of MCQs, the quality of explanation, and the diversity.

With our proposed methodology, the MCQs are generated in a balanced diversity of question types and difficulty levels aligned with different cognitive skill levels of Bloom's revised taxonomy. This makes the generated MCQs a suitable candidate as a question pool for reference to instructors. The CoVe technique has been adapted to reduce the false acceptance of questions with a wrong answer.

The proposed question types are suitable for a wide range of courses. The same methodology could potentially be applicable to these courses.

In addition to assisting educators in various teaching and learning activities with our automated generation of MCQs, our work contributes positively to AI-empowered education by offering readily actionable insights for educators and researchers to adapt in their classrooms and enlightening ideas for deeper investigation in the future.

APPENDIX A
PROMPT TEMPLATES

A. PROMPT TEMPLATE WITH USER-SPECIFIED SUBTOPICS

Prompt Template with User-specified Subtopics

You are a teacher of course { Introduction to Artificial Intelligence and Machine Learning }. A multiple-choice is a question with one correct answer and multiple distractors. The question and the correct answer are also called stem and key, respectively. Please generate { 25 } multiple-choice questions for the core topic { Searching } with subtopics: { Uniform-cost search, Greedy best-first search, and A* search }.

Prior Knowledge: { Calculus, probability and statistics, data structures and algorithms }

Course description: { This course aims to equip students with the main concepts, ideas and techniques of artificial intelligence (AI) including machine learning. It covers the basic building blocks of AI and its applications in different domains. This subject also aims to enable students to gain practical experiences in applying the techniques in AI to solve real-world problems. }

Questions should be formatted in the following styles:

(1) Each question should have { 5 } choices, where one and only one key, others are distractors. Label choices with letter A), B), C), ...
(2) Define graphs or neural networks at the beginning of the questions if they are needed.
(3) Correct answers should be given, along with detailed explanations including a step-by-step deduction of the answers.
(4) The output should be in json format including the keys ''question'', ''choices'', ''correctanswer'', ''explanation''.

Follow these generation guidelines:

(1) Understand the topic and subtopics: First, make sure you have a solid understanding of the core topic and subtopics and the related concepts, definitions, key properties, similarity and differences, and applications.
(2) Generate question with the following question types:
    a. Basic Definitions
    b. Key Characteristics
    c. Difference Identification
    d. Advantages and Disadvantages Comparison
    e. Example Scenarios
(3) Try to create distractors with common misconceptions.
(4) Try to create a few questions to test the creativity of students by either presenting a new, unexpected scenario and asking students to apply their learning creatively to solve a problem or meet a need, or posing counterfactual or speculative scenarios to spark imagination and requiring students to apply logic to unlikely situations.

(5) Balancing Questions: Ensure a balanced mix of easy, medium, and hard questions with different question types to cater to different levels of understanding.
(6) Avoid Ambiguity: Make sure each question has one and only one clear correct answer to avoid confusion.
(7) Neutral Distractor: Distractors should be different, and neither be obviously wrong nor give clues or hints to the correct answer.
(8) Check for Bias: Ensure that the questions are neutral and do not inadvertently suggest the correct answer through wording.

B. PROMPT TEMPLATE WITH SUBTOPIC AUTOMATION

Prompt Template with User-specified Subtopics

You are a teacher of course { Introduction to Artificial Intelligence and Machine Learning }. A multiple choice is a question with one correct answer and multiple distractors. The question and the correct answer are also called stem and key, respectively. Please generate { 25 } multiple-choice questions for the core topic { K-means Clustering}.

Prior Knowledge: { Calculus, probability and statistics, data structures and algorithms }

Course description: { This course aims to equip students with the main concepts, ideas and techniques of artificial intelligence (AI) including machine learning. It covers the basic building blocks of AI and its applications in different domains. This subject also aims to enable students to gain practical experiences in applying the techniques in AI to solve real-world problems. }

Questions should be formatted in the following styles:

(1) Each question should have { 5 } choices, where one and only one key, others are distractors. Label choices with letter A), B), C), ...
(2) Define graphs or neural networks at the beginning of the questions if they are needed.
(3) Correct answers should be given, along with detailed explanations including a step-by-step deduction of the answers.
(4) The output should be in json format including the keys "question", "choices", "correctanswer", "explanation".

Follow these generation guidelines:

(1) Understand the core topic: First, make sure you have a solid understanding of the core topic and the related concepts, definitions, key properties, similarity and differences, and applications.
(2) Generate question with the following question types:
   a. Basic Definitions
   b. Key Characteristics
   c. Difference Identification
   d. Advantages and Disadvantages Comparison
   e. Example Scenarios
(3) Try to create distractors with common misconceptions.
(4) Try to create a few questions to test the creativity of students by either presenting a new, unexpected scenario and asking students to apply their learning creatively to solve a problem or meet a need, or posing counterfactual or speculative scenarios to spark imagination and requiring students to apply logic to unlikely situations.
(5) Balancing Questions: Ensure a balanced mix of easy, medium, and hard questions with different question types to cater to different levels of understanding.
(6) Avoid Ambiguity: Make sure each question has one and only one clear correct answer to avoid confusion.
(7) Neutral Distractor: Distracters should be different, and neither be obviously wrong nor give clues or hints to the correct answer.
(8) Check for Bias: Ensure that the questions are neutral and do not inadvertently suggest the correct answer through wording.

APPENDIX B
RUBRICS FOR EVALUATION

1) Does the MCQ provide enough information to arrive at an answer?
   a) Yes, it provides enough information
   b) No, it doesn't provide enough information
   c) N/A or Don't know
2) Is there a correct answer listed in the options?
   Is the option marked "correct" actually correct?
   a) Yes, there is a correct answer and it is marked "correct"
   b) There is a correct answer but it is not marked "correct"
   c) There are multiple correct answers
   d) No, there is no correct answer
   e) N/A or Don't know
3) Are the options distinct from each other, ensuring they are unique choices?
   a) Yes, they are completely unique between each other
   b) Some choices are unique, some are too similar
   c) No, they are all too similar marking them repeated choices
   d) N/A or Don't know
4) Is the MCQ free from obviously wrong options?
   a) Yes, there are no obviously wrong options
   b) Yes, however the options give away the correct answer
   c) No, there are obviously wrong option(s)
   d) N/A or Don't know
5) Does correctly answering the MCQ help in mastering of the topics or subtopics?
   a) Yes, it does and the MCQ matches the topic or subtopics
   b) Yes, but the MCQ does not match well the topic or subtopics
   c) No, but the MCQ matches the topic or subtopics
   d) No, the MCQ does not match well the topic or subtopics
   e) N/A or Don't know

6) Does the explanation provided by LLM explain the correct answer of MCQ?
   a) Yes, it explains the correct option and why other options are wrong
   b) Yes, it explains the correct option only
   c) No, it neither explains the correct option nor the wrong options
   d) N/A or Don't know
7) What is the question type of the MCQ?
   a) Basic Definitions
   b) Key Characteristics
   c) Difference Identification
   d) Advantages and Disadvantages Com-parison
   e) Example Scenarios
   f) N/A or Don't know
8) What is the difficulty level of the MCQ?
   a) easy
   b) medium
   c) hard
   d) N/A or Don't know
9) What is the Revised Bloom's taxonomy level of the MCQ?
   a) Remembering
   b) Understanding
   c) Applying
   d) Evaluating
   e) Creating
   f) N/A or Don't know
10) Is this question a duplicate of another one in the topic?
   a) No, this question is not a duplicate of another one.
   b) Yes, this MCQ is a duplicate of another one, both questions address identical knowledge points from the same analytical perspective.
   c) N/A or Don't know
11) Is it a computation problem?
   a) Yes
   b) No

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CHI CHIU SO (Member, IEEE) received the B.Sc. degree majoring in mathematics and minoring in computer science and the Ph.D. degree in artificial intelligence from The University of Hong Kong. He is currently a Lecturer with the School of Professional Education and Executive Development, The Hong Kong Polytechnic University. His current research interests include exploring AI theories and their practical applications in areas like education and finance.



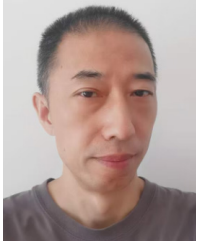
CHEUK HO LEE is currently pursuing the B.Sc. degree (Hons.) in computer science from The Hong Kong Polytechnic University. His research interests include advanced artificial intelligence (AI) in education technology and information retrieval. He is actively engaged in exploring tool-integrated reasoning (TIR). In addition to his work in the TIR, he researched in various sub-fields of AI, model merging, and machine learning optimization.



DAN YANG received the B.Sc. degree in educational technology from Qufu Normal University. She is currently pursuing the M.Sc. degree with the School of Education Science, Nanjing Normal University. Her research interests include student creativity, the application of artificial intelligence in education, and STEAM education.



ANTHONY WAI KEUNG LOH received the Bachelor of Science degree (Hons.) in mathematics and the Ph.D. degree in number theory from Imperial College, University of London. He possesses over 30 years of teaching experience in tertiary institutions and is currently the Director of Hong Kong Community College, The Hong Kong Polytechnic University. His research interests include analytic number theory, artificial intelligence, and noise engineering.



CHUEN KAN SZE received the Bachelor of Arts degree in mathematics and the Ph.D. degree in applied mathematics from The University of Hong Kong. He has been teaching at the School of Professional and Continuing Education, The University of Hong Kong, and conducting educational research at the School of Professional Education and Executive Development, The Hong Kong Polytechnic University. His current research interest includes AI-assisted education.